

PROGRAM 7

```
vedsk0831@DESKTOP-83FU7HB:~$ cd ~/ossp
mkdir -p program7
cd program7
vedsk0831@DESKTOP-83FU7HB:~/ossp/program7$ pwd
/home/vedsk0831/ossp/program7
vedsk0831@DESKTOP-83FU7HB:~/ossp/program7$ nano program7.c
vedsk0831@DESKTOP-83FU7HB:~/ossp/program7$ gcc -Wall -Wextra -g program7.c -o program7
vedsk0831@DESKTOP-83FU7HB:~/ossp/program7$ ls
program7  program7.c
vedsk0831@DESKTOP-83FU7HB:~/ossp/program7$ path
Command 'path' not found, did you mean:
  command 'xpath' from deb libxml-xpath-perl (1.48-1)
  command 'mpath' from deb libmodule-path-perl (0.19-3)
  command 'path6' from deb ipv6toolkit (2.0+ds.1-2build3)
  command 'patch' from deb patch (2.8-2build1)
  command 'yath' from deb libtest2-harness-perl (1.000161-1)
  command 'pat' from deb dist (1:3.5-236-1.1build1)
Try: sudo apt install <deb name>
vedsk0831@DESKTOP-83FU7HB:~/ossp/program7$ ./program7
=====
                OSSP PROGRAM 7
            waitpid() + PATH Searching
=====

Commands:
  path      - display PATH
  pathlist  - display PATH directories
  exit      - quit

shell7> path

PATH:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Window
sApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.4.7-
hotspot/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/
c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/Cloudflare/Cloudflare WARP:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs
/:/mnt/c/Users/HP/AppData/Local/Programs/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin:/mnt/c/Users/HP/AppData/Local/Microsoft/Windows
Apps:/mnt/c/Users/HP/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/HP/AppData/Local/Programs/DockerDesktop/resources/bin:
/mnt/c/Users/HP/AppData/Roaming/npm:/snap/bin
shell7> pathlist

PATH directories:
1. /usr/local/sbin
2. /usr/local/bin
3. /usr/sbin
4. /usr/bin
5. /sbin
6. /bin
7. /usr/games
8. /usr/local/games
9. /usr/lib/wsl/lib
10. /mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe
11. /mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.4.7-hotspot/bin
12. /mnt/c/WINDOWS/system32
13. /mnt/c/WINDOWS
14. /mnt/c/WINDOWS/System32/Wbem
15. /mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0/
16. /mnt/c/WINDOWS/System32/OpenSSH/
17. /mnt/c/Program Files/Cloudflare/Cloudflare WARP/
18. /mnt/c/Program Files/Git/cmd
19. /mnt/c/Program Files/nodejs/
20. /mnt/c/Users/HP/AppData/Local/Programs/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin
21. /mnt/c/Users/HP/AppData/Local/Microsoft/WindowsApps
22. /mnt/c/Users/HP/AppData/Local/Programs/Microsoft VS Code/bin
23. /mnt/c/Users/HP/AppData/Local/Programs/DockerDesktop/resources/bin
24. /mnt/c/Users/HP/AppData/Roaming/npm
```

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24. /mnt/c/Users/HP/AppData/Roaming/npm
25. /snap/bin
shell7> ls

Executable found:
/usr/bin/ls
Parent PID: 711
Child PID: 712
Parent waiting using waitpid()...

Child process started.
Child PID: 712
program7 program7.c
Child 712 exited normally.
Exit status: 0

shell7> ls -l

Executable found:
/usr/bin/ls
Parent PID: 711
Child PID: 713
Parent waiting using waitpid()...

Child process started.
Child PID: 713
total 32
-rwxr-xr-x 1 vedsk0831 vedsk0831 21944 Sep 17 05:02 program7
-rw-r--r-- 1 vedsk0831 vedsk0831 5949 Sep 17 05:02 program7.c
Child 713 exited normally.
Exit status: 0

shell7> pwd

```

```

Executable found:
/usr/bin/pwd
Parent PID: 711
Child PID: 714
Parent waiting using waitpid()...

Child process started.
Child PID: 714
/home/vedsk0831/ossdp/program7
Child 714 exited normally.
Exit status: 0

shell7> abcdef123
Command not found or not executable: abcdef123
shell7> exit
Program 7 terminated.
vedsk0831@DESKTOP-83FU7HB:~/ossdp/program7$ valgrind --leak-check=full --show-leak-kinds=all ./program7
==763== Memcheck, a memory error detector
==763== Copyright (C) 2002-2024, and GNU GPL'd, by Julian Seward et al.
==763== Using Valgrind-3.26.0 and LibVEX; rerun with -h for copyright info
==763== Command: ./program7
==763==
=====
OSSDP PROGRAM 7
waitpid() + PATH Searching
=====

Commands:
path      - display PATH
pathlist  - display PATH directories
exit      - quit

```

```

shell7> path

PATH:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/Window
sApps/MicrosoftCorporationII.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Eclipse Adoptium/jdk-25.0.4.7-
hotspot/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS:/mnt/c/WINDOWS/System32/Wbem:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/
c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/Cloudflare/Cloudflare WARP:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs
/./mnt/c/Users/HP/AppData/Local/Programs/Eclipse Adoptium/jdk-21.0.12.101-hotspot/bin:/mnt/c/Users/HP/AppData/Local/Microsoft/Windows
Apps:/mnt/c/Users/HP/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/HP/AppData/Local/Programs/DockerDesktop/resources/bin:
/mnt/c/Users/HP/AppData/Roaming/npm:/snap/bin
shell7> ls -l

Executable found:
/usr/bin/ls
Parent PID: 763
Child PID: 764
Parent waiting using waitpid()...

Child process started.
Child PID: 764
total 32
-rwxr-xr-x 1 vedsk0831 vedsk0831 21944 Sep 17 05:02 program7
-rw-r--r-- 1 vedsk0831 vedsk0831 5949 Sep 17 05:02 program7.c
Child 764 exited normally.
Exit status: 0

shell7> pwd

Executable found:
/usr/bin/pwd
Parent PID: 763
Child PID: 765

```

```

shell7> pwd

Executable found:
/usr/bin/pwd
Parent PID: 763
Child PID: 765
Parent waiting using waitpid()...

Child process started.
Child PID: 765
/home/vedsk0831/osspp/program7
Child 765 exited normally.
Exit status: 0

shell7> abcdef123
Command not found or not executable: abcdef123
shell7> exit
Program 7 terminated.
==763==
==763== HEAP SUMMARY:
==763==    in use at exit: 0 bytes in 0 blocks
==763==   total heap usage: 38 allocs, 38 frees, 5,802 bytes allocated
==763==
==763== All heap blocks were freed -- no leaks are possible
==763==
==763== For lists of detected and suppressed errors, rerun with: -s
==763== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
vedsk0831@DESKTOP-83FU7HB:~/osspp/program7$

```

C PROGRAM

#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <unistd.h>

#include <sys/types.h>

#include <sys/wait.h>

#define MAX_INPUT 1024

```
#define MAX_ARGS 100
```

```
/*  
 * Parse command and handle escaped characters.  
 *  
 * Example:  
 *  
 * echo hello world  
 *  
 * becomes:  
 * argv[0] = echo  
 * argv[1] = hello  
 * argv[2] = world  
 *  
 * echo hello\ world  
 *  
 * becomes:  
 * argv[0] = echo  
 * argv[1] = hello world  
 */
```

```
int parse_command(char *input, char *args[])  
{  
    int argc = 0;  
    int i = 0;  
  
    while (input[i] != '\0' && argc < MAX_ARGS - 1)  
    {  
        /* Skip spaces */
```

```
while (input[i] == ' ' || input[i] == '\t')
{
    i++;
}
```

```
if (input[i] == '\0')
{
    break;
}
```

```
/* Start of a new argument */
args[argc++] = &input[i];
```

```
int write_pos = i;
```

```
while (input[i] != '\0')
{
    /*
     * Backslash escape
     *
     * hello\ world
     *
     * becomes
     *
     * hello world
     */
    if (input[i] == '\\' && input[i + 1] != '\0')
    {
```

```

    i++;

    input[write_pos++] = input[i++];
}

/*
 * Normal character
 */
else if (input[i] != ' ' && input[i] != '\t')
{
    input[write_pos++] = input[i++];
}

/*
 * Normal space means
 * end of current argument.
 */
else
{
    i++;
    break;
}

}

/* Terminate argument */
input[write_pos] = '\0';
}

```

```

/* exec requires NULL at the end */
args[argc] = NULL;

return argc;
}

int main()
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("=====\n");
    printf("    OSSP PROGRAM 6\n");
    printf("  Escape Sequences & Process Execution\n");
    printf("=====\n");

    printf("\nExamples:\n");
    printf(" ls -l\n");
    printf(" echo hello\n");
    printf(" echo hello\\ world\n");
    printf(" pwd\n");
    printf(" date\n");

    printf("\nType 'exit' to quit.\n\n");

    while (1)

```

```
{  
    printf("shell6> ");  
    fflush(stdout);  
  
    /*  
     * Read command  
     */  
    if (fgets(input, sizeof(input), stdin) == NULL)  
    {  
        break;  
    }  
  
    /*  
     * Remove newline  
     */  
    input[strcspn(input, "\n")] = '\0';  
  
    /*  
     * Ignore empty input  
     */  
    if (strlen(input) == 0)  
    {  
        continue;  
    }  
  
    /*  
     * Exit  
     */  
}
```

```
if (strcmp(input, "exit") == 0)
{
    break;
}

/*
 * Parse command
 */
int argc = parse_command(input, args);

if (argc == 0)
{
    continue;
}

/*
 * Display arguments
 */
printf("\nParsed arguments:\n");

for (int i = 0; i < argc; i++)
{
    printf("  argv[%d] = [%s]\n", i, args[i]);
}

/*
 * Create child process
```

```

*/

pid_t pid = fork();

if (pid < 0)
{
    perror("fork");
    continue;
}

/*
 * CHILD PROCESS
 */
if (pid == 0)
{
    printf("\nChild process created.\n");
    printf("Child PID: %d\n", getpid());

    /*
     * Execute command.
     *
     * execvp searches PATH automatically.
     */
    execvp(args[0], args);

    /*
     * Reached only if execvp fails.
     */

```

```

    perror("execvp");

    exit(EXIT_FAILURE);
}

/*
 * PARENT PROCESS
 */
else
{
    printf("Parent PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);

    printf("Parent waiting for child...\n");

    int status;

    /*
     * Wait for child to finish
     */
    waitpid(pid, &status, 0);

    /*
     * Check child status
     */
    if (WIFEXITED(status))
    {

```

```
    printf(
        "Child exited with status: %d\n",
        WEXITSTATUS(status)
    );
}
else
{
    printf("Child terminated abnormally.\n");
}
}

printf("\n");
}

printf("Program 6 terminated.\n");

return 0;
}
```